Study-specific information

What is the study about?

In this study, named *Wearanize+*, we will use several wearable devices to record sleep. We will use the data to evaluate the quality of the sleep readouts and design algorithms to compare and improve their quality.

What is expected of you?

We will first explain the study procedure, the rules and regulations (detailed in the attached documents), and your rights at the beginning of the study. Then we will ask you to wear these wearable devices that collect various physiological and sleep data from different places of the body:

Device name	Recording modalities	In plain words
SOMNOscreen Plus	EEG[1], EMG[2], EOG[3], ECG[4], & actigraphy	has electrodes that record brainwaves, eye movements, heart rate, & muscle activity
Zmax	EEG, pulse, & actigraphy	a headband that records brainwaves, heart rate, & movement
mSafety (Sony)	Pulse & actigraphy	a wristband that records heart rate, pulse, & movement
Gmeter	EEG & NIRS[5]	a headband that records brainwaves & brain-waste clearance
Empatica	Pulse, EDA[6], temperature, & actigraphy	a wristband that records heart rate, pulse, electrical conductance, & movement
ActivPAL	Actigraphy	a leg-patch that records movement

[1]EEG is a measure of brain activity. Please see the "Information About Wearable Sleep EEG" document for details.

[2]EMG is a measure of electrical activities within the muscle. This data will be measured using three electrodes placed on the chin (and the throat) and will be used to detect movement and REM sleep periods.

[3]EOG is a measure of eye movement. This data will be collected using two electrodes placed on the side of both eyes and will be used to monitor eye movement during sleep.

[4]ECG is a measure of the electrical activity of the heart. This data will be measured using two electrodes (one placed near the right shoulder and the other on the lower left chest) and will be used to record the heart activity throughout the night.

[5]NIRS will be used to monitor the flow of cerebrospinal fluid within the brain during sleep. The cerebrospinal fluid cushions the brain, providing essential protection, nutrients, and clearance. The headband has one LED source and two detectors to detect its flow during sleep. This data will help us to investigate the relation between sleep stages and glymphatic clearance.

[6]EDA measures the changes in the conductivity of the skin caused by sweating. The wristband has a built-in sensor to measure EDA of the contact area.

A researcher will set up these devices on various parts of your body and show and explain how to use them. All of these measurements are non-invasive. Although most devices will start recording as soon as you wear them, you must turn some on before going to sleep. You will also need to perform a few jumps and left-right (LR) eye movements before and after sleep. A researcher will show you how to do these activities. You will sleep in your home/residence and should aim for normal, uninterrupted sleep during regular sleep times. Except for your sleep and physiological data, no data from your surroundings will be collected. Data collected by these devices will give us insights into your sleep physiology. You can request a sleep report containing details of your sleep of the recorded night. We will further ask you to fill out the following questionnaires to assess your subjective sleep, dreams, and other physical and emotional characteristics:

Questionnaire name	Regarding
Pittsburgh Sleep Quality Index (PSQI)	Subjective sleep quality over the past month
Mannheim Dream Questionnaire (MADRE)	Dream experience over the last year
Evening/Morning Sleep Questionnaire (DLQ)	Sleep experience of recording night
Patient Health Questionnaire (PHQ-9)	Mood assessment over the past two weeks
Comfort Questionnaire	Your comfort in wearing the devices

You will receive 50 € for participating in the study. The session will take place in the late afternoon or evening (close to your usual sleep time) and take around 45–60 minutes. You can fill in the questionnaires at home. We will collect the devices from you the day after. If some devices fail to record your sleep (or their data quality is not good), we might ask you to wear the devices for another night.

What will happen to your data?
Your participation in this study is confidential. All research data will be stored on Radboud University's secure servers, according to the university's protocol. This protocol is in line with the General Data Protection Regulation (GDPR).

Questions regarding the project, to your participation, or the recorded data:
Your participation in the study voluntary. You may say no at any time. You do not have to answer questions you would rather not answer, and you can stop your participation and withdraw your consent at any time during the study. You do not have to indicate why you are stopping. If you want to participate in this study, we will ask you to sign a consent form. Your written consent indicates that you have understood the information and agree to participate in the study. You can also have your research data and personal data deleted up to two weeks after participation, by sending an email to the principal investigator **Martin Dresler (martin.dresler@donders.ru.nl)**.